AMERICAN ACADEMY OF OPHTHALMOLOGY®

Grip Strength and Age-Related Ocular Diseases: Insights from Observational, Mendelian Randomization, and Mediation Analyses

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Objective: We aimed to examine the cross sectional and causal associations of grip strength with cataract, glaucoma, diabetic retinopathy (DR), and age-related macular degeneration (AMD) and probe the underlying mechanisms by evaluating the mediating role of metabolomic alterations.

Design: Cross sectional study.

Subjects: A total of 307 796 UK Biobank participants with grip strength and covariates data available.

Methods: Logistic regression models were used to evaluate the associations between grip strength and age-related ocular diseases. Two-sample Mendelian randomization analyses were conducted to assess the causality. Metabolic biomarkers from plasma samples were measured through nuclear magnetic resonance (n = 152 376), and principal component (PC) analysis was implemented to identify metabolic patterns (PC1–PC8). The mediation effects of both metabolic biomarkers and metabolic patterns were examined.

Main Outcome Measures: The prevalence of age-related ocular diseases.

Results: Compared with the highest tertile of grip strength, the lowest tertile had a higher prevalence of cataract (odds ratio [OR], 1.31; 95% confidence interval [CI], 1.24–1.39), glaucoma (OR, 1.23; 95% CI, 1.13–1.33), and DR (OR, 2.80; 95% CI, 2.32–3.38). Genetic-associated elevated grip strength of at least 1 hand was associated with a lower risk of developing cataracts, DR, and AMD. Mediation analyses showed metabolic patterns, characterized by altered lipids and omega-3 polyunsaturated fatty acids (PUFAs) decrement (i.e., PC2 and PC8), significantly mediated the association of grip strength with cataract and DR.

Conclusions: Weaker grip strength is associated with cataracts, glaucoma, and DR. Metabolomic alterations, especially disrupted lipid metabolism and omega-3 PUFA decrement, serve to be the critical mediators.

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Grip strength is one of the pragmatic measures of skeletal muscle function and is proposed to be an age-related marker.[1–3] It has been reported to be associated with multiple age-related disorders, such as dementia and Parkinson disease.[4–7] Given the critical role of skeletal muscle function in age-related disorders, a possible link between grip strength and age-related ocular diseases may also exist.

To this end, several past studies investigated the associations between grip strength and age-related ocular diseases.[8–11] However, several knowledge gaps remain. First, the existing results were inconsistent across different studies. For instance, the Beaver Dam Eye Study reported the associations of grip strength with age-related macular degeneration (AMD) in men,[8] whereas in the Ural Eye and Medical Study, no such association was observed.[9] Further, investigation on the causal associations between grip strength and age-related ocular diseases is lacking.

Moreover, the mechanisms underlying the associations between grip strength and age-related ocular diseases have rarely been explored. The pivotal role of skeletal muscle in whole-body energy metabolism suggests that alterations in grip strength may lead to metabolomic profile changes.[12] Previous studies have shown that weak grip strength was associated with abnormal levels of metabolites and the incidence of metabolic diseases.[13–16] On this note, metabolic derangement was also reported in the pathogenesis of age-related ocular diseases.[17–19] Nevertheless, to date, no study has examined the mediating role of metabolic alterations in these associations.

In this study, utilizing data from the UK Biobank with a large sample size of >300 000 participants, we aimed to

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https://doi.org/10.1016/j.xops.2025.100831
ISSN 2666-9145/25

comprehensively evaluate the associations between grip strength and age-related ocular diseases. We first used the logistic regression model to explore the associations of grip strength with cataracts, glaucoma, diabetic retinopathy (DR), and AMD. Next, we conducted Mendelian randomization (MR) to assess the causality of the aforementioned associations. Finally, we evaluated the mediating role of metabolomic alterations in the associations between grip strength and age-related ocular diseases.

Methods

Study Design and Participants

This large-scale population-based study was based on the data from the UK Biobank, in which >500 000 participants aged 37 to 73 years were enrolled between 2006 and 2010 and underwent a series of baseline testing items at an assessment center across the United Kingdom with extensive information collected.[20] The UK Biobank study adhered to the Declaration of Helsinki and received ethics approval from the National Information Governance Board for Health and Social Care and North West Multicenter Research Ethics Committee (11/NW/0382) with written informed consent obtained from all participants. The study protocols and other relevant information are available online (https://www.ukbiobank.ac.uk/). This work was conducted under the application number 93118.

We excluded individuals who missed any data on grip strength or covariates (n = 192 031). Those with a grip strength of >2.5 standard deviations (SDs) away from the median were also removed (n = 2532).[15] Eventually, 307 796 participants were included in the final analysis, and the flowchart of participant selection was shown in Figure S1 (available at https://www.ophthalmologyscience.org).

Assessment of Grip Strength

Grip strength was measured using a Jamar J00150 hydraulic hand dynamometer. The strength of both hands was measured in turn, and the average value expressed in absolute units (kg) was used for the subsequent analysis. Grip strength was then categorized into tertiles based on their age and sex to ensure the comparability,[4] and the cutoff values were described in Table S1 (available at https://www.ophthalmologyscience.org).

Age-Related Ocular Diseases

The cases of interest in this study were 4 age-related ocular diseases,[21] namely cataract, glaucoma, DR, and AMD. Participants were considered cases if they self-reported any diagnosis of ocular diseases through the touchscreen questionnaire. Additionally, we also included cases from the hospital inpatient-sourced data based on the International Classification of Diseases codes at any point before and up to 1 year after the baseline assessment.[22–24] Detailed information on definitions and codes was provided in Table S2 (available at https://www.ophthalmologyscience.org).

Covariates

Sociodemographic factors including age, sex, and ethnicity were self-reported by the participant during the initial assessment center visit, and the Townsend deprivation index was calculated based on the participant's output area. Lifestyle factors considered as confounders included alcohol intake frequency[25] categorized as daily or almost daily, 3 or 4 times a week, once or twice a week, 1 to 3 times a week, special occasions only, and never; smoking status[26] categorized as never, former, and current; and total metabolic equivalent task minutes per week for all activities[27] that was derived based on the International Physical Activity Questionnaire guidelines.[28] The following 5 components of metabolic syndrome were also included as potential confounders.[29,30] History of hypertension was obtained through a verbal interview by a trained nurse and the codes used to define the hypertension referred to a previous study.[31] Body mass index was calculated based on the weight and standing height measured by trained nurses and was categorized into underweight (<18.5 kg/m^2), normal weight (18.5 to <25.0 kg/m^2), overweight (25.0 to <30.0 kg/m^2), and obese (≥30.0 kg/m^2).[4] The sample size of underweight people was rather small, and they were thus combined into the normal weight group in the analysis.[3] Concentration of high-density lipoprotein cholesterol, triglycerides, and glycated hemoglobin were measured in the blood sample collected at the basement. Corneal-compensated intraocular pressure for both eyes was measured using the Reichert Ocular Response Analyzer, and the mean value was used in further sensitivity analysis.

Nuclear Magnetic Resonance Metabolomics Measurement

High-throughput nuclear magnetic resonance-based metabolic biomarker profiling platform developed by Nightingale Health Ltd was used to measure 249 metabolic biomarkers from randomly selected ethylenediaminetetraacetic acid plasma samples. Detailed information on experimental procedures has been described previously.[32] The biomarkers included lipoprotein lipids in 14 subclasses (concentration and proportion), fatty acids, and various low-molecular-weight metabolites, such as amino acids, ketone bodies, and glycolysis metabolites quantified in molar concentration units. In our study, data on metabolic biomarkers were available in 152 376 participants.

Statistical Analysis

Baseline characteristics by grip strength were presented as means and SDs for continuous variables, and frequencies and percentages for categorical variables. Comparisons between groups were performed using the Kruskal–Wallis or chi-square test according to the type of data.

The logistic regression model was applied to examine the associations between grip strength and age-related ocular diseases, and the results were expressed as odds ratios (ORs) with 95% confidence intervals (CIs). Analyses were adjusted for age, age-squared (age^2), sex, ethnicity, the Townsend deprivation index, alcohol intake frequency, smoking status, total metabolic equivalent task, body mass index category, triglycerides, high-density lipoprotein cholesterol, glycated hemoglobin, and history of hypertension. Variance inflation factors for all coefficients were estimated to assess the collinearity between variables, and all obtained values were below 3.0, which therefore did not support the existence of severe collinearity problems.[33] Furthermore, a restricted cubic spline was implemented to determine the nonlinear relationships between grip strength and age-related ocular diseases. Subsequent subgroup analyses were conducted stratified by age (<65 years, ≥65 years) and sex (female, male). An interaction term of stratum factor and grip strength was added into the model to identify potential interaction effects. Finally, 2 sensitivity analyses were performed to assess the robustness of our findings: (1) exploring the longitudinal association between grip strength and age-related ocular diseases based on the hospital

inpatient data; and (2) analyzing the association between grip strength and glaucoma with further adjustment of intraocular pressure in a subgroup of 67 608 participants with available intraocular pressure data.

Mendelian randomization analysis was performed to estimate the causal relationships between grip strength and age-related ocular diseases. The summary statistics of grip strength of right and left hands were obtained from genome-wide association studies conducted by Neale laboratory (http://www.nealelab.is/), which covered >330 000 individuals of European ancestry. Single nucleotide polymorphisms (SNPs) as genetic instruments of grip strength were selected at the threshold $P < 5 \times 10^{-8}$, and linkage disequilibrium clumping ($r^2 < 0.01$, window size = 10 000 kilobases) was conducted. We further used RadialMR to calculate the modified Q test and discard the outliers with a nominal significance level of 0.05.[34] Genetic associations with cataract were obtained from a large published genome-wide association studies meta-analysis (n = 585 243);[35] associations with glaucoma were obtained from the International Glaucoma Genetics Consortium (n = 216 257);[36] and associations with DR were obtained from the FinnGen consortium (n = 216 666); associations with AMD were obtained from the International AMD Genomics Consortium (n = 33 976).[37] The beta and standard errors for variants of AMD are not available in its summary statistics and were calculated according to a prior MR study involving AMD.[37] Inverse-variance weighted methods with random effects were used in our primary analysis to estimate the causal effect and address the heterogeneity issue.[38,39] Several other methods, including MR-Egger,[40] weighted median,[41] and weighted mode,[42] were also performed. Leave-one-out analysis was conducted to examine whether the causal association was driven by a SNP. Mendelian randomization-Egger intercept test and Cochran Q test were used to assess potential horizontal pleiotropy and heterogeneity, respectively.[40,43] All MR analyses were performed by R package "TwoSampleMR."[44]

In mediation analysis, all 249 metabolic biomarkers were processed by ranked-based inverse normal transformation and then were standardized to a mean of 0 and an SD of 1.[45] The investigation consisted of 2 steps. First, the mediation effect of each metabolic biomarker was estimated separately to fully screen potential mediators and find profound biomarkers with large mediation effects. The regression-based approach was utilized to estimate the natural indirect effect (i.e., mediation effect) and mediated proportion.[46] Second, the principal component (PC) analysis was conducted on 249 metabolic biomarkers to extract metabolic patterns with several PCs generated.[45] The mediation effects of metabolic patterns, indicated by PCs, were then investigated using a regression-based approach as discussed earlier.[46] The false discover rate (FDR) obtained by the Benjamini–Hochberg method was used to correct for multiple testing. All mediation analyses were performed by the R package "CMAverse."[47]

The statistical tests were 2-sided, and a P value <0.05 was considered statistically significant. All statistical analyses were performed using R 4.3.0 (R Foundation for Statistical Computing).

Results

Baseline Characteristics

A total of 307 796 UK Biobank participants were included in the final analysis (Fig S1). Baseline characteristics by grip strength were described in Table 3. Individuals with a lower grip strength were more likely to be older, male, non-British, more deprived, and a current smoker and had a relatively lower alcohol intake frequency and less physical activity. They also tended to have a higher body mass index, triglycerides and glycated hemoglobin, a history of hypertension, and a lower high-density lipoprotein cholesterol. Baseline characteristics of subgroup for analyses related to metabolic biomarkers (n = 152 376) were shown in Table S4 (available at https://www.ophthalmologyscience.org).

Observational Associations between Grip Strength and Age-Related Ocular Diseases

Observational associations between tertiles of grip strength and age-related ocular diseases were shown in Figure 2. Compared with those in the highest tertile of grip strength, individuals in the lowest tertile had higher prevalence of cataract (OR, 1.31; 95% CI, 1.24–1.39; $P < 0.001$), glaucoma (OR, 1.23; 95% CI, 1.13–1.33; $P < 0.001$), and DR (OR, 2.80; 95% CI, 2.32–3.38; $P < 0.001$) in the fully adjusted models. Individuals in the middle tertile of grip strength also had higher prevalence of cataract (OR, 1.10; 95% CI, 1.04–1.16; $P = 0.001$) and DR (OR, 1.57; 95% CI, 1.27–1.94; $P < 0.001$). However, no association was identified between grip strength and prevalent AMD (OR_{middle}, 1.05; 95% CI, 0.86–1.27, $P = 0.623$; OR_{lowest}, 1.16; 95% CI, 0.96–1.41; $P = 0.124$).

When grip strength was treated as a continuous variable, restricted cubic spline showed a nonlinear association between grip strength and cataract (Fig 3A; nonlinear $P = 0.016$). The risk of cataract increased gradually because grip strength decreased until around 30 kg and increased markedly afterward (i.e., <30 kg; Fig. 3A). No other significant nonlinear patterns were observed (Fig 3B–D). In subgroup analysis stratified by age and sex (Table S5, available at https://www.ophthalmologyscience.org), the effect of weaker grip strength on cataract was significantly higher in the older subgroup (≥65 years) than in the younger subgroup (<65 years) (OR_{middle}, 1.15; 95% CI, 1.06–1.25 vs. OR_{middle}, 1.10; 95% CI, 1.02–1.19, P for interaction = 0.018). No other significant interactions were observed. Finally, sensitivity analysis showed similar results to the primary analysis (Table S6, available at https://www.ophthalmologyscience.org). Of note, the longitudinal analysis also showed an association between grip strength and incident AMD (hazard ratio 1.21; 95% CI, 1.14–1.30; $P < 0.001$).

Causal Relationships between Grip Strength and Age-Related Ocular Diseases

Causal relationships between grip strength and age-related ocular diseases were then investigated through 2-sample MR analysis (Table S7, available at https://www.ophthalmologyscience.org). Under the inverse-variance weighted model, genetic-associated elevated grip strength in at least 1

Table 3. Baseline Characteristics of the UK Biobank Participants by Grip Strength

	Overall n = 307 796	Highest n = 108 319	Middle n = 103 719	Lowest n = 95 758	*P* value
Sociodemographics					
Age, mean (SD), yrs	56.3 (8.1)	56.0 (8.3)	56.2 (8.1)	56.7 (7.9)	<0.001
Sex, No. (%)					<0.001
Male	148 368 (48.2)	52 039 (48.0)	48 053 (46.3)	47 482 (49.6)	
Female	159 428 (51.8)	56 280 (52.0)	55 666 (53.7)	48 276 (50.4)	
Ethnicity, No. (%)					<0.001
British	275 210 (89.4)	98 366 (90.8)	93 575 (90.2)	83 269 (87.0)	
Others	32 586 (10.6)	9953 (9.2)	10 144 (9.8)	12 489 (13.0)	
Townsend deprivation index, mean (SD)	−1.5 (3.0)	−1.7 (2.9)	−1.5 (3.0)	−1.1 (3.2)	<0.001
Lifestyle factors					
Smoking status, No. (%)					<0.001
Never	168 514 (54.7)	58 545 (54.0)	57 266 (55.2)	52 703 (55.0)	
Former	108 295 (35.2)	39 286 (36.3)	36 208 (34.9)	32 801 (34.3)	
Current	30 987 (10.1)	10 488 (9.7)	10 245 (9.9)	10 254 (10.7)	
Alcohol intake, No. (%)					<0.001
Daily or almost daily	66 389 (21.6)	25 204 (23.3)	22 538 (21.7)	18 647 (19.5)	
3 or 4 times a week	74 610 (24.2)	27 712 (25.6)	25 674 (24.8)	21 224 (22.2)	
Once or twice a week	79 420 (25.8)	27 979 (25.8)	26 938 (26.0)	24 503 (25.6)	
1–3 times a month	33 583 (10.9)	11 514 (10.6)	11 479 (11.1)	10 590 (11.1)	
Special occasions only	31 987 (10.4)	9956 (9.2)	10 430 (10.1)	11 601 (12.1)	
Never	21 807 (7.1)	5954 (5.5)	6660 (6.4)	9193 (9.6)	
Total MET, mean (SD), min/wk	2647.3 (2650.6)	2793.3 (2703.1)	2666.9 (2637.7)	2460.9 (2592.9)	<0.001
Health status					
BMI categories, No. (%)					<0.001
<25.0 kg/m^2	104 518 (34.0)	35 166 (32.5)	37 001 (35.7)	32 351 (33.8)	
25.0 to <30.0 kg/m^2	132 306 (43.0)	48 140 (44.4)	44 273 (42.7)	39 893 (41.7)	
≥30.0 kg/m^2	70 972 (23.1)	25 013 (23.1)	22 445 (21.6)	23 514 (24.6)	
Triglycerides, mean (SD), mmol/l	1.7 (1.0)	1.7 (1.0)	1.7 (1.0)	1.8 (1.0)	<0.001
HDL-C, mean (SD), mmol/l	1.5 (0.4)	1.5 (0.4)	1.5 (0.4)	1.4 (0.4)	<0.001
HbA1c, mean (SD), mmol/mol	35.9 (6.6)	35.5 (5.7)	35.7 (6.5)	36.6 (7.5)	<0.001
Hypertension, No. (%)	78 496 (25.5)	26 252(24.2)	25 217 (24.3)	27 027 (28.2)	<0.001

BMI = body mass index; HbA1c = glycated hemoglobin; HDL-C = high-density lipoprotein cholesterol; MET = metabolic equivalent of task; SD = standard deviation.

hand was associated with lower risk of developing cataract (OR_{right}, 0.80; 95% CI, 0.72–0.89; $P < 0.001$; OR_{left}, 0.78; 95% CI, 0.65–0.93, $P = 0.006$), glaucoma (OR_{right}, 0.76; 95% CI, 0.59–0.98; $P = 0.032$), DR (OR_{left}, 0.74; 95% CI, 0.58–0.94; $P = 0.015$), and AMD (OR_{right}, 0.20; 95% CI, 0.11–0.38; $P < 0.001$; OR_{left}, 0.14; 95% CI, 0.08–0.27; $P < 0.001$). These results were further supported by at least one of the other MR models (weighted median, weighted mode, and MR-Egger) except for glaucoma (Table S7). The scatterplots of SNP effects on grip strength versus age-related ocular diseases are presented in Figure S4 (available at https://www.ophthalmologyscience.org). Forest plots of individual SNP MR-estimated effect sizes are presented in Figure S5 (available at https://www.ophthalmologyscience.org). Leave-one-out analysis indicated that no SNP drove the causal association between grip strength and cataract, DR, or AMD, but this was not the case for glaucoma (Fig S6, available at https://www.ophthalmologyscience.org). Mendelian randomization-Egger intercept test showed no evidence for the existence of horizontal pleiotropy, and estimates from Cochran Q statistic for heterogeneity were presented in Table S8 (available at https://www.ophthalmologyscience.org).

Plasma Metabolic Biomarkers and Patterns Mediated Associations between Grip Strength and Age-Related Ocular Diseases

Lower grip strength was correlated with altered levels of 202 (81.1%) metabolic biomarkers after FDR correction, indicating that altered grip strength was strongly associated with metabolomic changes (Fig. S7A and Table S9, available at https://www.ophthalmologyscience.org). Among these relationships, it was prominent that weaker grip strength was associated with higher levels of glycoprotein acetyls (GlycA, $R = 0.122$, $P_{FDR} < 0.001$) and larger ratio of monounsaturated to total fatty acids (MUFA%, $R = 0.094$, $P_{FDR} < 0.001$). It was also remarkable that weaker grip strength was associated with lower levels of albumin ($R = -0.125$, $P_{FDR} < 0.001$), creatine ($R = -0.113$, $P_{FDR} < 0.001$), and citrate ($R = -0.086$, $P_{FDR} < 0.001$). The associations between grip strength and age-related ocular diseases are shown in Figure S7B and Table S10 (available at https://www.ophthalmologyscience.org). The number of metabolic biomarkers associated with DR was the largest (n = 222, 89.2%), followed by cataract (n = 172, 69.1%).

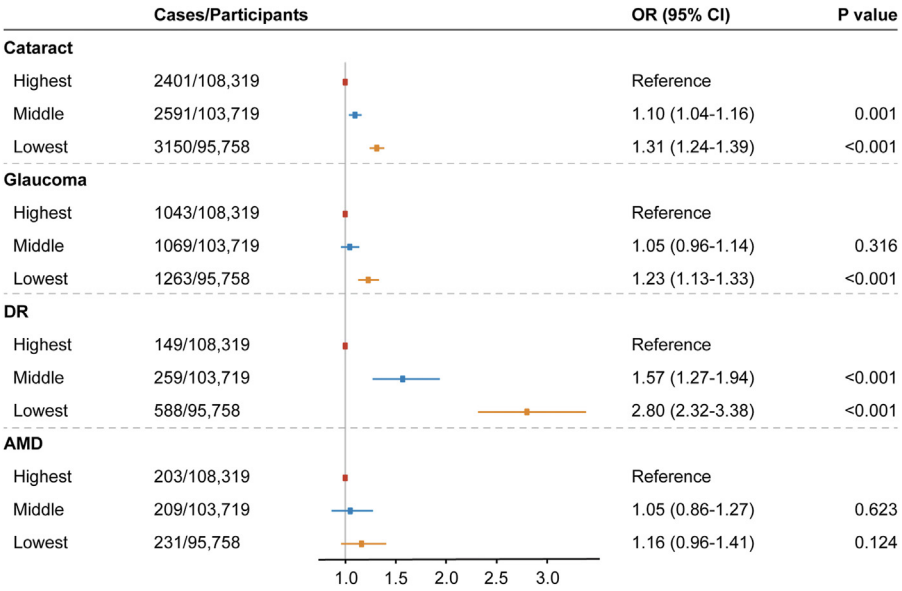


Figure 2. Associations between grip strength and age-related ocular diseases. The ORs and 95% CIs for cataract, glaucoma, DR, or AMD were estimated using logistic regression models adjusted for age, age-squared, sex, ethnicity, the Townsend deprivation index, alcohol intake frequency, smoking status, total MET, BMI categories, triglycerides, HDL-C, HbA1c, and history of hypertension. AMD = age-related macular degeneration; BMI = body mass index; CI = confidence interval; DR = diabetic retinopathy; HbA1c = glycated hemoglobin; HDL-C = high-density lipoprotein cholesterol; MET = metabolic equivalent of task; OR = odds ratio.

No metabolic biomarkers were associated with glaucoma or AMD.

In separate mediation analysis, 169 metabolic biomarkers mediated the association between grip strength and DR, accounting for 67.9% of total biomarkers. This was followed by cataract, with 118 biomarkers (47.4%) (Fig 8A). Metabolic biomarkers with the top 5 mediation effects are shown in Figure 8B. Detailed information on mediation effects can be found in Table S11 (available at https://www.ophthalmologyscience.org).

Principal component analysis was further conducted on all 249 metabolic biomarkers, and the top 8 PCs (cumulatively explaining 88.10% of the overall variance) were selected for further analysis after considering both the interpretability and variance explained. The loadings and the main characteristics of these 8 PCs are provided in Figure S9 (available at https://www.ophthalmologyscience.org) and Tables S12 and S13 (available at https://www.ophthalmologyscience.org). Associations between grip strength and PCs are shown in Figure 10A and Table S14 (available at https://www.ophthalmologyscience.org), and associations between PCs and age-related ocular diseases are shown in Figure 10B and Table S15 (available at https://www.ophthalmologyscience.org). The mediation effects of the 8 PCs were then investigated. Remarkably, PC2 and PC8 had the highest mediation effects on both grip strength-DR and grip strength-cataract associations, and their mediated proportions were 6.97% and 2.33% for grip strength-DR association, respectively, and 1.28% and 1.16% for grip strength-cataract association, respectively (Fig 10C and Table S16, available at https://www.ophthalmologyscience.org). Additionally, PC5 mediated 1.04% of the grip strength-DR association. No PCs mediated associations between grip strength and glaucoma.

Discussion

In this large-scale population-based study of 307 796 UK Biobank participants, we systematically evaluated the associations between grip strength and age-related ocular diseases. We demonstrated that weak grip strength was associated with a higher prevalence of cataract, glaucoma, and DR. Further, MR analyses supported a causal association of grip strength with cataract, DR, and AMD, although the association with glaucoma is less robust. We also probed the underlying mechanisms by revealing the mediating role of metabolomic alterations in the associations of grip strength with cataract and DR.

Previous studies on associations between grip strength and ocular diseases have been conducted in relatively smaller sample sizes and have yielded inconsistent results. In the Ural Eye and Medical Study (n = 5381), higher grip strength was found to be associated with a lower prevalence of DR ($\beta = -0.03$, $P = 0.007$), whereas no association was observed between grip strength and cataract (nuclear, cortical, subcapsular posterior), primary open-angle glaucoma, or AMD.[9] Although in the Study of Health in Pomerania (n = 1173), the absence of association between grip strength and AMD was also concluded,[10] the Beaver Dam Eye Study (n = 2962) showed associations of grip strength with both early (OR/10 kg decrease 1.28; 95% CI, 1.08−1.52) and late AMD (OR, 1.55; 95% CI, 1.02−2.36) in men but not in women.[8] Moreover, the

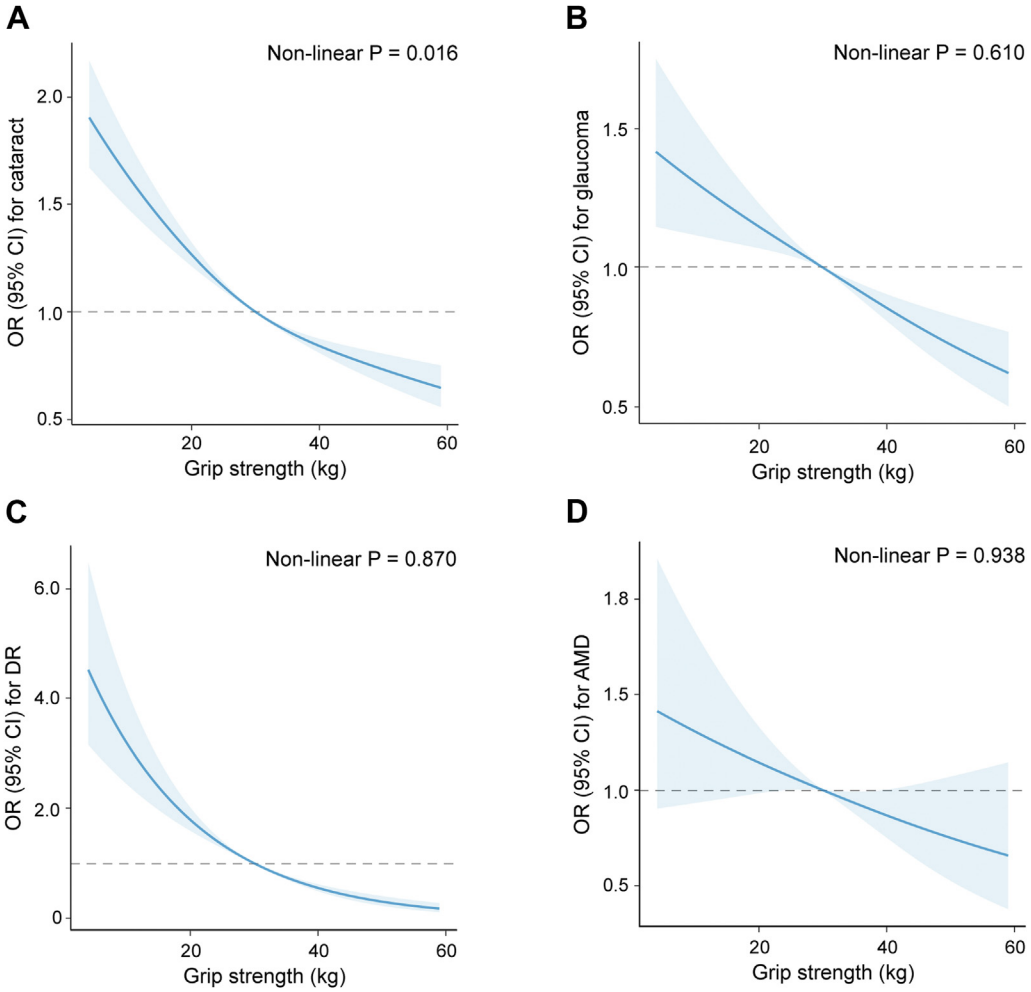


Figure 3. Nonlinear associations between grip strength and age-related ocular diseases. Data of (**A**) cataract, (**B**) glaucoma, (**C**) DR, and (**D**) AMD were presented as ORs with the 95% CIs shown as shading area. Nonlinear associations were investigated using restricted cubic splines fit in logistic regression models adjusted for age, age-squared, sex, ethnicity, the Townsend deprivation index, alcohol intake frequency, smoking status, total MET, BMI categories, triglycerides, HDL-C, HbA1c, and history of hypertension. AMD = age-related macular degeneration; BMI = body mass index; CI = confidence interval; DR = diabetic retinopathy; HbA1c = glycated hemoglobin; HDL-C = high-density lipoprotein cholesterol; MET = metabolic equivalent of task; OR = odds ratio.

Beaver Dam Eye Study (n = 2370) also found the opposite conclusion to the Ural Eye and Medical Study that cortical cataract was correlated with weaker grip strength in men (estimate −1.60; 95% CI, −2.90 to −0.20).[11] Additionally, these analyses were limited by the lack of data on potential confounders. Some important covariates, such as ethnicity and components of metabolic syndrome, failed to be adjusted in these studies.

With extensive information on multiple confounders and a large sample size in the UK Biobank, our population-based study demonstrated that weak grip strength was associated with cataract, glaucoma, and DR. Further longitudinal and MR analyses supported associations of grip strength with cataract, glaucoma, DR, and AMD. The failure to establish an association between grip strength and AMD in the cross sectional analysis may be because of low awareness for AMD in participants. In our data set, only 643 individuals self-reported or were hospitalized for AMD before baseline, which may possibly be an underestimate. A community-based study found only a third of patients with AMD were aware of being affected by AMD.[48] It was also reported that approximately 25.0% of eyes having AMD were deemed to be normal by primary eye care physicians.[49] Therefore, studies based on a more accurate AMD ascertainment method were required in the future to elaborate the actual association between grip strength and AMD.

We further interrogated the mediating role of metabolomic alterations to gain a deeper understanding of underlying mechanisms. Our 2-step analyses first consistently ascertained the mediating role of disrupted lipid metabolism. In mediation analysis on separate metabolic biomarkers, the lipid proportions in some subclasses of very low-density lipoprotein, intermediate-density lipoprotein, and low-density lipoprotein had relatively large mediation effects in the associations of grip strength with cataract and DR. The

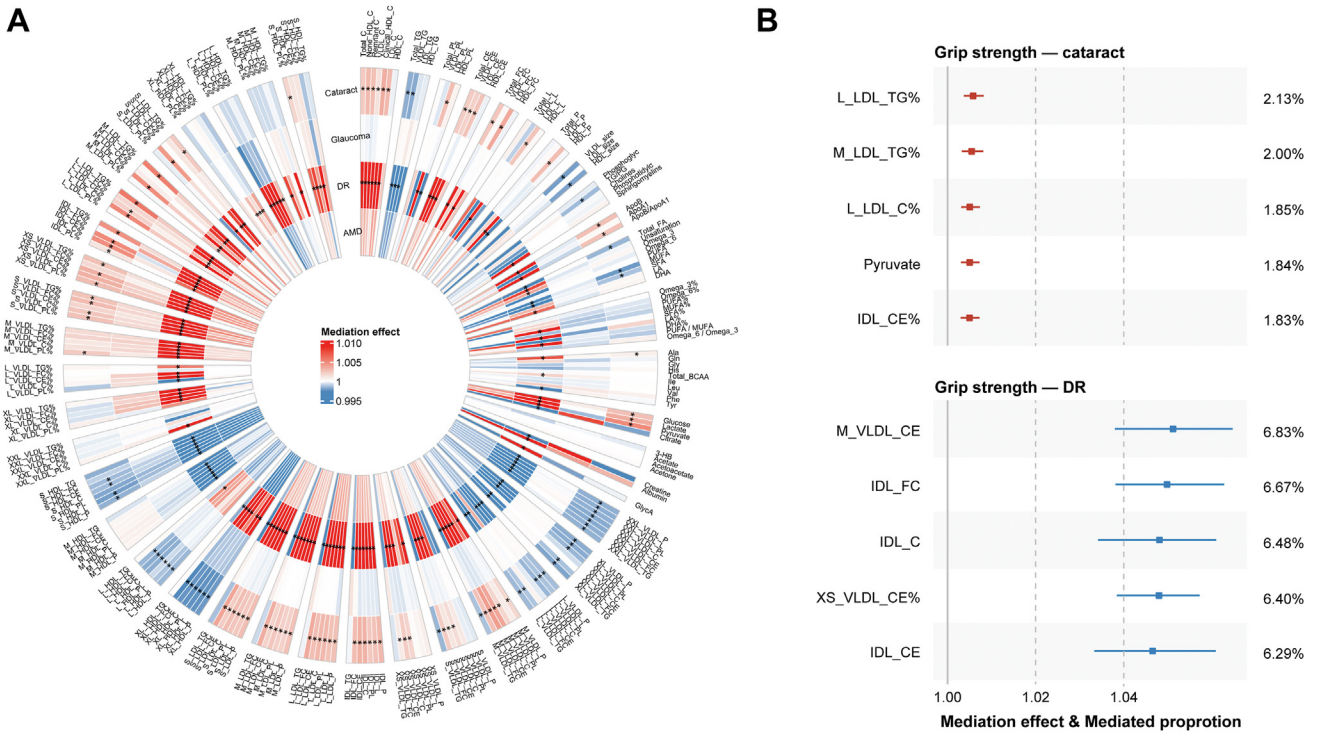


Figure 8. The mediating role of metabolic biomarkers in the associations between grip strength and age-related ocular diseases. **A,** Natural indirect effect (i.e., mediation effect) of each metabolic biomarker that reported on the odds ratio scale. The significant mediation effect of $P_{FDR} < 0.05$ was indicated by an asterisk (*). **B,** Metabolic biomarkers with the top 5 mediation effects for associations of grip strength with cataract or DR. Mediated proportions were displayed on the right side of the figure. AMD = age-related macular degeneration; DR = diabetic retinopathy; FDR = false discovery rate.

mediating role of abnormal lipid metabolism was further supported by the analysis of metabolic patterns, as we identified PC2, whose score was mainly associated with various lipids especially those in very low-density lipoprotein, intermediate-density lipoprotein, and low-density lipoportein (see Table S13, available at https://www.ophthalmologyscience.org), profoundly mediated the effects of weaker grip strength on cataract and DR. These findings were partly supported by the fact that disrupted lipid metabolism is a common feature of muscle aging and ocular diseases.[50–54] Further, the mediating role of polyunsaturated fatty acids (PUFAs) decrement, especially omega-3 PUFAs decrement, in the grip strength-DR association stood out. The PUFA decrement and monounsaturated fatty acids increment owned large mediation effects in separate mediation analysis. Likewise, a metabolic pattern indicated by PC8, whose score was mainly associated with high levels of omega-3 PUFAs, docosahexaenoic acid, degree of unsaturation, and low levels of omega-6 to omega-3 PUFAs ratio, also significantly mediated grip strength-DR association. Omega-3 PUFAs possess antioxidative, anti-inflammatory, and neuroprotective properties[55] and play a key role in retinal integrity and health.[56,57] More importantly, a randomized controlled trial confirmed supplementation of dietary intake of omega-3 PUFAs could lower the risk of DR[58] and also improve the upper/lower extremity muscle strength in patients with sarcopenia,[59] suggesting this approach may serve as a potential intervention strategy in preventing DR resulting from skeletal muscle dysfunction.

Our study has several strengths. First, our study is based on a large sample size and adjusts for multiple important confounders. Second, we identify the causal relationships between grip strength and age-related ocular diseases by MR analysis. Third, our work is the first attempt to establish the relationship between grip strength, metabolomic alterations, and age-related ocular diseases and demonstrates the mediating role of both metabolic biomarkers and patterns. Our work is also subjected to some limitations. First, the UK Biobank is subjected to a low response rate and health volunteer bias[60] and may not be representative of the general UK population. Second, our study mainly includes White participants and the findings we obtain should be replicated in other ethnicities. Third, the cross-sectional design of our study precludes a causal inference to our results. Fourth, the ascertainment of age-related ocular diseases through self-report can be subject to recall and misclassification bias and a low diagnosis rate in the primary eye care. Although we also merge cases from the hospital inpatient data, this approach may only capture serious cases requiring hospitalization and may not be very specific to eye diseases. Fifth, there exists a problem of sample overlap in our MR analysis regarding the relationship between grip strength and cataract, which may lead to biased causal estimates. Finally, altered levels of some metabolites may be the

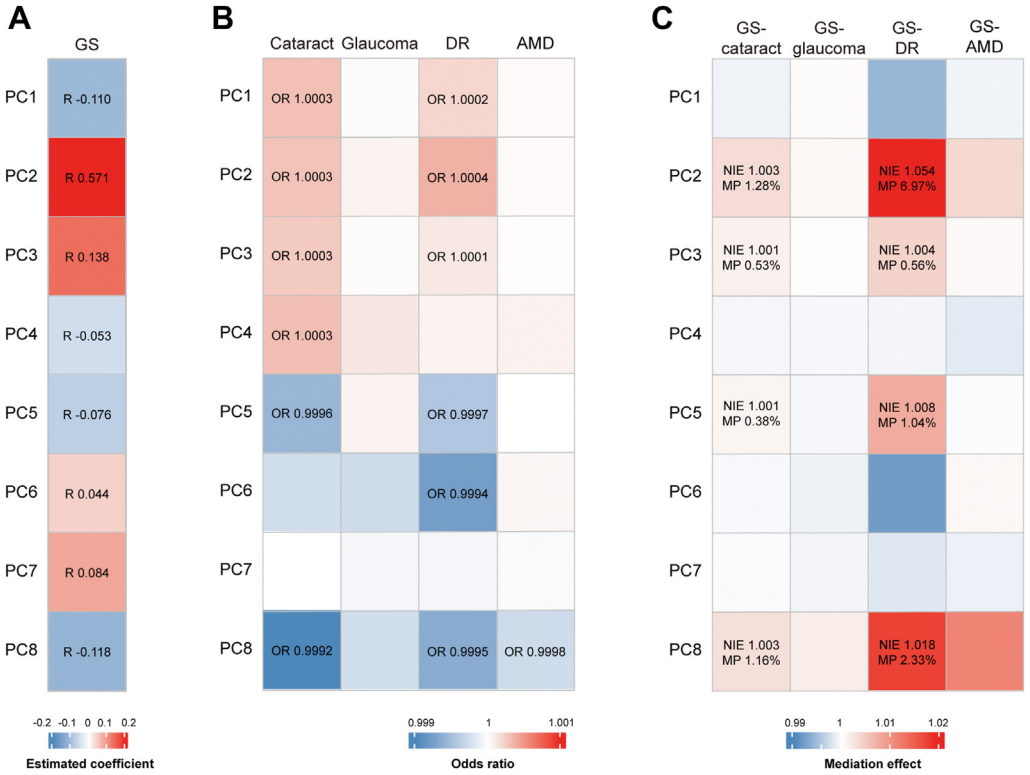


Figure 10. Associations between grip strength and age-related ocular diseases with metabolic principal components as potential mediators. **A**, Associations between grip strength and metabolic patterns (indicated by PCs). The correlation coefficient (*R* value) for the lowest tertile of grip strength was estimated using multiple linear regression models, with the highest tertile of grip strength as the reference group. **B,** Associations between metabolic patterns (indicated by PCs) and age-related ocular diseases. The OR of age-related ocular disease was estimated using logistic regression models. All regression models in (**A**) and (**B**) were adjusted for age, sex, ethnicity, the Townsend deprivation index, alcohol intake frequency, smoking status, total MET, BMI categories, history of hypertension. **C,** Natural indirect effects (i.e., mediation effects), and mediated proportions of PCs as potential mediators in the associations between grip strength and age-related ocular diseases. Only significant *R* values, ORs, or mediation effects with $P_{FDR} < 0.05$ were shown in the grids. AMD = age-related macular degeneration; BMI = body mass index; DR = diabetic retinopathy; FDR = false discovery rate; GS = grip strength; MET = metabolic equivalent of task; OR = odds ratio; PC = principal component.

contributors rather than consequences of grip strength decline. The hypothetic causal framework adopted in mediation analysis, that is, grip strength dysfunction results in metabolomic alterations which finally lead to ocular diseases, may oversimplify the relationships between the 3 in complex biological systems.

Grip strength is a pragmatic measurement and a modifiable factor that can be enhanced through exercise. Our study provides preliminary evidence that grip strength may be useful in the detection and prevention of age-related ocular diseases. Nevertheless, more studies are required to rigorously appraise the practical value of grip strength in the management of age-related ocular diseases.

In conclusion, our study demonstrates weaker grip strength is associated with a higher prevalence of cataract, glaucoma, and DR. Mendelian randomization analysis supports the causality between grip strength and cataract, DR, and AMD. Metabolomic alterations, especially disrupted lipid metabolism and omega-3 PUFA decrement, play a mediating role in the associations of grip strength with cataract and DR. Overall, these findings consistently highlight the association between grip strength and age-related ocular diseases and underscore the potential of prevention and intervention strategies oriented by grip strength and associated critical metabolites for age-related ocular diseases.

Acknowledgments

The authors are grateful to all the participants and researchers from the UK Biobank.

Footnotes and Disclosures

Originally received: January 19, 2025.
Final revision: May 12, 2025.
Accepted: May 15, 2025.
Available online: May 21, 2025. Manuscript no.: XOPS-D-25-00043R2.

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Disclosure(s):

All authors have completed and submitted the ICMJE disclosures form.

The authors have no proprietary or commercial interest in any materials discussed in this article.

Supported by research grants from the National Key Research and Development Program of China (2022YFC2502800), the National Natural Science Foundation of China (82271069, 82371040, 82122017, 81870642, 81970780, 81470613 and 81670835), Science and Technology Innovation Action Plan of Shanghai Science and Technology Commission (23Y11909800), Outstanding Youth Medical Talents of Shanghai "Rising Stars of Medical Talents" Youth Development Program, and Shanghai Municipal Health Commission Project (2024ZZ1025 and 20244Z0015). The sponsor or funding organization had no role in the design or conduct of this research.

HUMAN SUBJECTS: Human subjects were included in this study. The UK Biobank study adhered to the Declaration of Helsinki and received ethics approval from the National Information Governance Board for Health and Social Care and North West Multicenter Research Ethics Committee (11/NW/0382) with written informed consent obtained from all participants.

No animal subjects were used in this study.

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Analysis and interpretation: Chen, Guo, Meng

Obtained funding: Zhu

Overall responsibility: Chen

Abbreviations and Acronyms:

AMD = age-related macular degeneration; **CI** = confidence interval; **DR** = diabetic retinopathy; **FDR** = false discover rate; **MR** = Mendelian randomization; **OR** = odds ratio; **PUFA** = polyunsaturated fatty acid; **PC** = principal component; **SD** = standard deviation; **SNP** = single nucleotide polymorphism.

Key words:

Age-related macular degeneration, Cataract, Diabetic retinopathy, Glaucoma, Grip strength.

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